

Article

Automating Product Design and Fabrication Within the Furniture Industry

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Abstract

Furniture is an integral part of daily life. Its comfort and usability are key factors that define its success. In recent years, there has been increasing demand for applications that drive businesses toward Industry 4.0. These applications aim to improve productivity through greater automation in both 3D modeling and fabrication processes. This research aims to develop a Computer Aided Design (CAD) platform that automates the design and manufacturing of furniture. The platform is based on visual programming using Grasshopper 3D™ and provides a solid foundation for processing different geometric shapes. These shapes can be customized according to the user's preferences. The platform's innovation lies in its ability to process complex geometries with a fully automated algorithm. Once the initial parameters are set, the algorithm generates the results. The input data includes an initial geometry, which can be highly complex. Additionally, a set of construction parameters is introduced, leading to multiple alternative design solutions based on the same initial geometry. The designer and user can select their final choice, and all resulting design and manufacturing outcomes are automatically generated. These outcomes include 3D part models, 3D assembly files, Bill of Materials, G-code for CNC machining, and nesting capabilities for improved material efficiency. The platform ensures high-quality performance. The results of the study show that the platform successfully works with different geometries. Moreover, the study is significant as the Industry 4.0 transformation moves toward more automated design processes.

Keywords: furniture design; computational design; product design; parametric design; automatic fabrication



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1. Introduction

Holistic design plays a key role when developing products, considering a variety of aspects and downstream applications. In particular within the furniture industry, instead of using traditional techniques, computational design offers a great deal of opportunities. Automating the whole process results in increased productivity, while considering the different design and fabrication processes involved [1].

The furniture industry uses mainly traditional processes for designing its products and thus there is a strong demand to use state-of-the-art computer-aided design (CAD) tools to be able to substantially shorten the product development time and offer products based on non-conventional geometries. In this way, a new approach to the designers' role is needed. The designer involves the user from the beginning of the project, while at the same

time incorporating visual or textual programming coding within advanced CAD systems. This approach can facilitate a customized automated design approach, thus drastically improving its efficiency.

2. State of the Art

Furniture automated design is a fast-growing area of research. Research tends to mainly focus on two alternative foci. The first deals with the design and manufacturing automation based on using a general-purpose CAD system, making use of its API (Application Programming Interface) capabilities, i.e., the use of SolidworksTM 2021 and its textual programming capability, leading to integrating both the design and the manufacturing of a coffee table. Another direction is the use of visual programming tools, i.e., Grasshopper 3DTM (part of the RhinoTM 3D CAD system), which can incorporate significantly more complex furniture geometries [2]. Computational design using a variety of 3D geometries was proposed for implementation in the design of furniture for open spaces and parks by Bianconi and Filippucci. In this way, parametric design technology was combined with architectural prototypes and digital fabrication methods, with an aim to produce sustainable products. The use of recyclable materials of marine plywood and ethylene-vinyl acetate has contributed towards the application of sustainability principles. At the same time, nesting tools and CNC milling code preparation have assisted in substantially decreasing the fabrication and assembling process, together with the reduction in the material waste [3]. The term “parametric design”, described above, refers to computer-aided design that is modified based on some parameters. Parametric design is included within computational design, which is the broader concept of using algorithms and computational processes. In other words, computational design is not limited to parameters only as it can include optimization, artificial intelligence, genetic algorithms, and many others. In contrast, parametric design requires the use of variables that will be used as parameters to modify the final geometry [4–6]. Implementing customization in furniture design and fabrication, Manavis et al. achieved significantly more efficient procedures. Both the design and fabrication performance improved the overall product efficiency and at the same time, involving the end user in the design process offered significantly more customer satisfaction. The end user was able to define a number of alternative geometrical characteristics and material selection, while at the same time 3D modeling and fabrication files were becoming available for direct use [7]. Ma et al. integrated a series of technologies, i.e., computer graphics, topology optimization, and digital fabrication, and offered an excellent opportunity to promote the initial furniture design while optimizing the final products. Newly developed chairs and tables can effectively be developed following the proposed framework, while efficiency and cost reduction were identified. The unusual geometries proposed were fabricated using additive manufacturing technology as the only means to do so. The presentation of the case studies provided a solid basis for using the proposed methodology within the furniture industry [8]. The importance of advanced CAD systems within the office furniture design was recognized by Hu et al. and the parametric design principles were used with an aim to analyze the design requirements. The modular aspects and the structural characteristics were selected as the main features to be optimized when modern office furniture was needed. The main calculation formulas were revealed in order to finalize the proposed furniture modeling and provide increased customer satisfaction [9].

Additive manufacturing can play a key role within the furniture manufacturing process. A series of specially designed joint components were used to replace the wooden joint components that were difficult and time-consuming to build. This approach was proposed by Nicolau et al. and can help designers to expand their innovation and design inspiration without limits, while novel furniture can be designed with a variety of joint

geometries. The development of a 3D printed connector that could be used for joining three chair components offered additional advantages from the geometry point of view. At the same time, several evaluation methods were used. The material examined was a combination of reinforced polylactic acid (PLA) and fiberglass (20 wt. %) filament. Its performance proved significantly increased [10]. Bodenschatz and Rosenthal proposed the use of liquid deposition modeling based on a binder of carboxymethyl cellulose with wood flour and this was effectively used to manufacture table frames. A novel additive manufacturing device was developed to accommodate the new physical material properties determined in this research [11]. Alternative 3D printing technologies have been tested for producing lower cost components in a variety of industries. Wood powder-based additive manufacturing methods offer additional fabrication opportunities. This is a widely available renewable material and offers lower manufacturing costs as the material can be used combined with synthetic or natural binders, thus strongly promoting sustainable design. It is an area that needs far more exploration and research, but the initial findings have led to positive results. Environmental sustainability, material recycling, and cost reduction can be achieved through technological advancements in 3D printing. Several studies are investigating the printing of products and furniture using materials that use wood powder. In this way, small batches can be produced, reducing material waste and thus the overall cost [12–17]. The ratio of polymers to wood is a factor that significantly affects the tensile strength of a 3D printed product. This is the reason that a great deal of adhesive was used to provide a ready-to-be-used material [18–20]. At the same time, the shape and size of the wood grains play important roles in the properties of the material produced [21]. Corresponding studies investigated the effect of printing parameters such as nozzle temperature, speed, layer thickness, etc., on the mechanical properties of 3D-printed specimens when the FFF (Fused Filament Fabrication) method is followed [22–27]. In addition, researchers have carried out comparative studies by varying the shape of the internal filling and the filling percentage [28–31].

Computational design can integrate the design and manufacturing processes under a unique platform via CAD-based programming. Through visual and/or textual programming algorithms, the designer can automate the 3D modeling required as a basis for downstream applications. The key advantages of this method over traditional design are the reduced time needed via the achievement of process automation, the ease of designing complex geometries, and the development of multiple design solutions. Grasshopper 3D™, which is an extension of Rhino 3D™ Version 7, is one of the most popular tools for product design engineering. In recent years, there has been an increase in the integration of Python™ and artificial intelligence tools in the field of computational product design, thus making the whole process more versatile while altering the role of both the designer and the user as well. Designers should acquire programming skills and the users should participate in a variety of design stages, thus substantially increasing their satisfaction and commitment [32–35].

Computational product design can certainly be extensively used in the furniture industry. Techniques such as origami, that are supported by mathematical algorithms and equations, can be integrated into the design process in order to automate the proposed design and fabrication solutions [36]. At the same time, computational design has been used in urban architecture in which parametrically designed structures based on wood were presented, e.g., wooden gazebos and outdoor benches. In some cases, the use of sophisticated mathematical modeling is considered an asset for increased creativity [37]. During the parametric design of furniture, the designer needs to incorporate the principles of creative thinking via algorithmic design and develop highly complex forms, thus improving the creativity of the proposed solutions [38].

Waste reduction during furniture creation can be substantially improved via computational design applications. In essence, embedded algorithms can calculate in advance a series of alternative solutions and optimize the material usage, leading to minimum material requirements and waste [39]. This can be achieved despite the complex geometries and forms that are modeled and proposed for fabrication. Cutting many thin surfaces and bending wood through 3D schematic cutting can be easily implemented and parameterized through an algorithm. Grasshopper3D™ is a very popular tool for incorporating CAD-based visual programming into a variety of product design and fabrication cases [40–43]. The use of artificial intelligence and machine learning complement digital furniture 3D design tools, together with topology optimization algorithms that lead to the use of additive manufacturing processes for the fabrication of the resulting 3D geometries (e.g., Autodesk™ Fusion™ 360, PTC Creo™, CATIA™, 3D Rhinoceros™ and Ansys™ Discovery™) [44–46]. Material selection in the furniture industry plays a key role not only from the cost point of view but for the ease of manufacturing and assembly. Wood is considered to be a very popular choice, given all the possibilities for recycled and raw material, i.e., medium density fiberboard (MDF), particle board (DSP), fiberboard (DVP), plywood, rattan. Alternative materials that can be used include metal alloys for increased resistance to damage, glass for giving lightness to the environment it is used in, stone for its high resistance to mechanical damage, and plastic for its low weight and low cost. In addition, natural or artificial leather and textiles are used for supporting furniture quality [47]. There are a number of studies focusing on the development of integrated workflows using parametric tools and additive manufacturing processes. García-Domínguez et al. proposed a continuum framework as a methodology that combines parametric design, optimization, and 3D printing [48]. Similarly, Kontovourkis and Tryfonos combined parametric design with robotic prosthetic manufacturing for clay printing in a continuum framework. In this continuum framework, the automatic translation of design parameters to a robotic arm was performed [49]. Another example that highlights the current trend of research studies towards integrated systems was presented by Taher et al. They presented an integrated framework that starts with the product design itself and ends in the fabrication of structural elements through clay prosthetic manufacturing [50].

This paper aims to present a comprehensive design and fabrication platform based on visual programming within a CAD system. It generates a wide range of furniture alternatives using various geometric forms. The platform is built on the idea that the designer takes on dual roles: both a design engineer and a CAD programming expert. The user can input their own form—whether modeled in another 3D CAD system or 3D-scanned if necessary—and set up the initial manufacturing parameters. The platform then automatically designs the final product using a slicing fabrication approach. The outputs include 3D CAD models, 3D sub/assembly files, a bill of materials (BOM), prototyping files (stl, step, vectors), computer-aided manufacturing codes (CAM), nesting, and rendering tools. The main contributions of this research are the fully automated process, which requires no user intervention, and the flexibility to select the initial geometric form without restrictions. The proposed design framework serves as an integrated design and manufacturing platform. It not only automates the entire design and fabrication process but also supports complex geometries and involves the user from the start of the furniture design phase. This approach ultimately leads to higher customer satisfaction within the furniture industry.

3. Methodology

Figure 1 illustrates the framework proposed in this research. The goal was to automate furniture design based on non-conventional geometries, primarily for pieces to be fabricated using wood-related materials, such as plywood. The design engineer's role was to integrate

both the furniture design and fabrication processes, along with the necessary programming skills, to automate the entire workflow. In this system, the user can input an unusual geometry along with other fabrication parameters, such as the number of slices, slice thickness, setscrew and spacer properties, and default values for nesting and manufacturing processes. A series of calculations then results in a complete furniture design based on slicing fabrication principles, with all product details displayed on the screen. The system generates a set of outputs supporting the technical aspects of fabrication, including computer-aided manufacturing (CAM) code, nesting, bill of materials, rendering, and prototyping files. Rhino 3D™ Version 7 and its visual programming language, Grasshopper 3D™, were used to facilitate and automate the entire process. This design framework is tailored to wooden furniture products and can be adapted to develop a range of specialized applications.

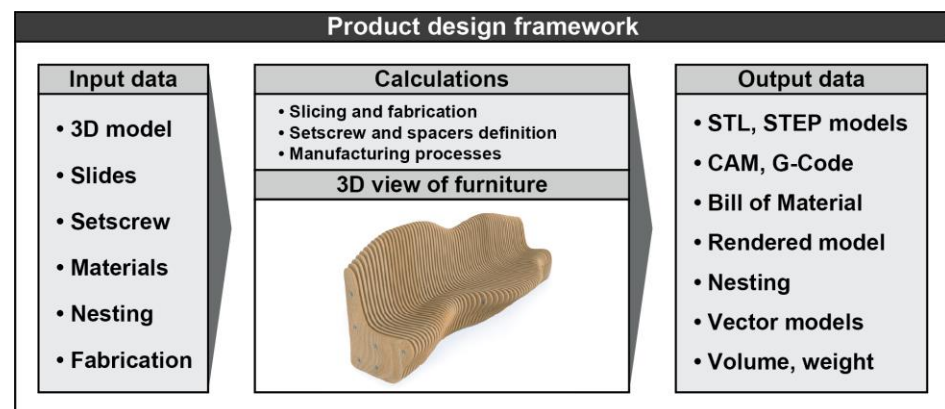


Figure 1. The proposed furniture design framework.

3.1. Algorithm Development and Parameterization

A visual programming language (VPL) was used to create the design platform. More specifically, the development of the algorithm was implemented in Grasshopper 3D™, the visual programming language included in 3D Rhino™, because it is a general purpose CAD system and it can support complex 3D geometries manipulations. Although the main part of the algorithm was developed with Grasshopper 3D™, several additional pieces of code were developed using textual programming via Python™ version 2.7. The use of Python™ was introduced to simplify several pieces of the code while optimizing them. In essence, they created a coherent result organized into several Python™ nodes. In addition, although visual programming helps in the rapid development of the code, it lags in complex mathematical problems compared to Python™. In other words, there is a strong demand for integrating textual and visual programming tools in order to create appropriate automated applications.

Figure 2 graphically presents the operation of each set of nodes used. Each node performs the input, processing, and output of various data. The proposed algorithm is divided into four main stages:

- Stage A (input data, volume division): Initially, the algorithm requires input from the user. The first piece of information is the definition of the 3D geometry that the furniture will be based on. Then, additional data, i.e., number of slices, setscrew radius, guiding points, wood panels thickness, are defined. After the data insertion process, the algorithm leads to dividing the 3D furniture model into slices based on the user's requirements.
- Stage B (setscrew direction, setscrew holes): The algorithm defines the direction of the setscrews according to the orientation of the slices created. A hole is introduced in

each slice at the points at which each setscrew passes through. These holes use the dimensions of the setscrews with the corresponding tolerances for correct and easy assembly. The designer has the option to manually change the paths created if needed.

- Stage C (create screw nuts, create spacers and setscrews, assemble furniture): The algorithm redesigns each part of the furniture in detail, i.e., the threads on the nuts are defined and spacers and setscrews are designed with additional details. Based on the redesigned parts, the 3D modeling of the complete assembly is performed with the parametric characteristics of the furniture.
- Stage D (nesting, BOM and export final model, G-Code, set materials and render): The slices are converted from 3D to 2D format and positioned on the plane for the application of the nesting method. The 2D format of the slices is a vector-based geometry on which the G-Code required for the fabrication is created. In addition to the G-Code, a series of exported data is supported by the algorithm, such as 3D CAD models, nesting information, bill of materials (BOM), and export of the final 3D model. Finally, the materials and textures that have been defined from the beginning by the user are applied to create photorealistic renders.

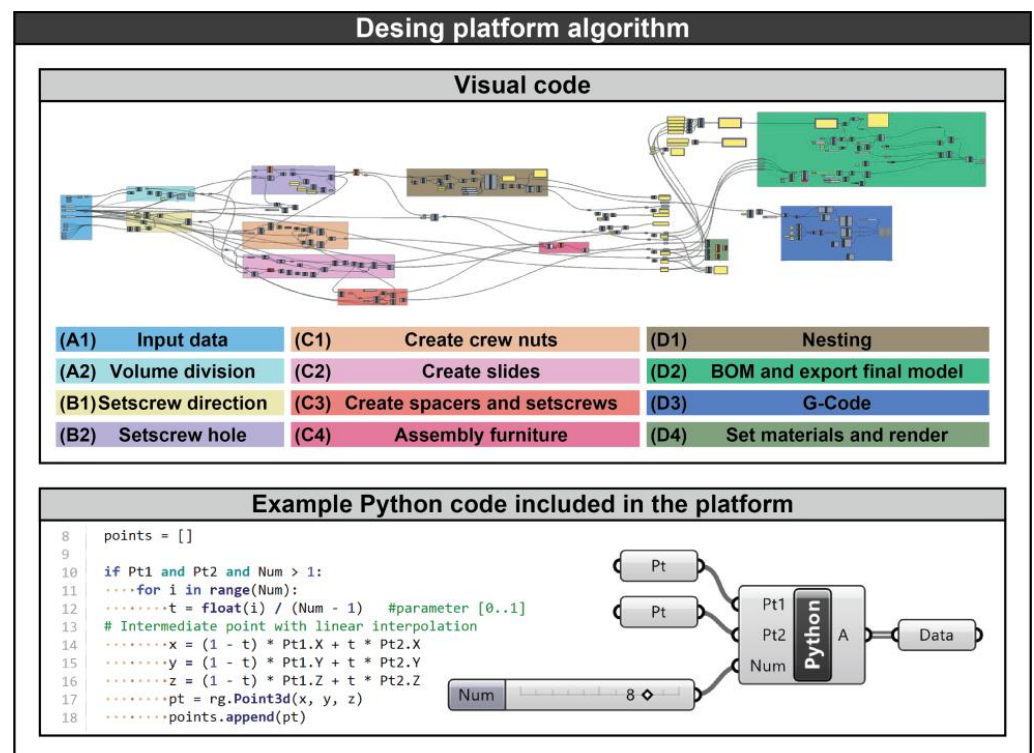


Figure 2. Design platform algorithm presentation.

3.2. Input Data

The user is able to provide an unusual geometry to be used as the basis for the final furniture design. A 3D CAD file is transferred to Rhino 3D™ (i.e., stl, obj, step). Based on material selection, several properties were set:

- Material selection, e.g., plywood with different coloring, thickness in mm, material's panel overall dimensions (i.e., 2.50 m × 1.25 m).
- Definition of the number of slices used for the complete product: This definition was related to the number and size of the spacers designed.
- Selection of the setscrew's positions: All the properties that relate to the direction of the setscrew, their size (diameter), and their ending are set in this step.

- Default properties are considered for the nesting procedure and all fabrication setup used (algorithms, tooling, manufacturing parameters, etc.).

Figure 3 illustrates how the automated furniture design platform operates when a simplified geometry is introduced. The system automatically selects the side of the volume that is perpendicular to its longest length (the initial geometry can also be sliced based on a user-selected plane, if needed). Next, setscrews are inserted automatically, though the designer can modify them if necessary. Spacers are used to fill the gaps between the slices, and finally, the end pieces are designed and positioned automatically.

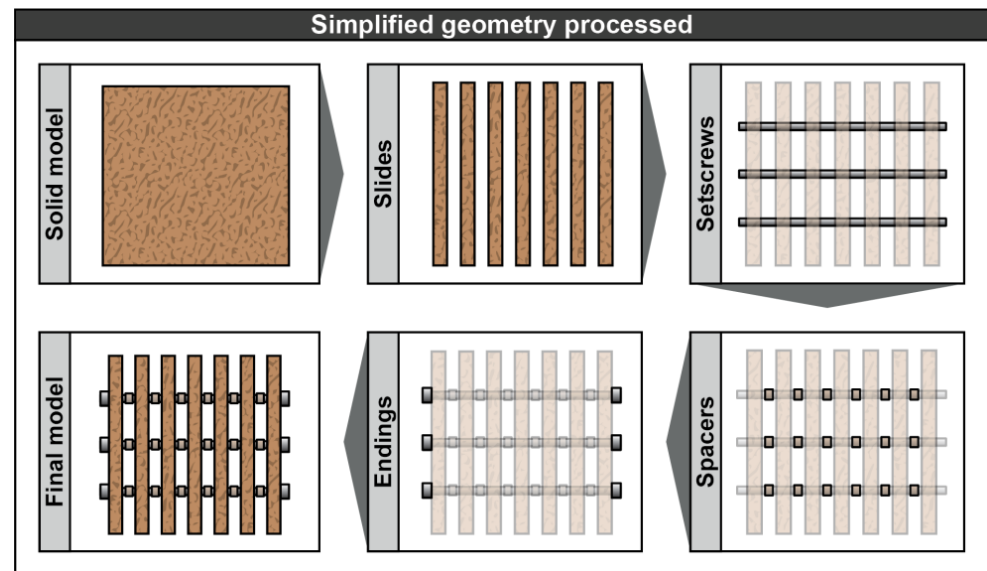


Figure 3. Simplified geometry processed by the implemented platform.

3.3. Design Automation

When the input geometry has an internal gap, the algorithm halts the development of the setscrews. The process resumes once the furniture geometry is complete. This interruption ensures that the setscrews remain hidden when the geometric gap is present (see Figure 4). The placement of setscrews is defined automatically by the algorithm. It calculates the placement based on side edge points that are distant from the slice edges. The number of setscrews is determined by the area of the slices. If the automatic placement does not yield the desired result, the user can manually edit and optimize both the number and positioning of the setscrews.

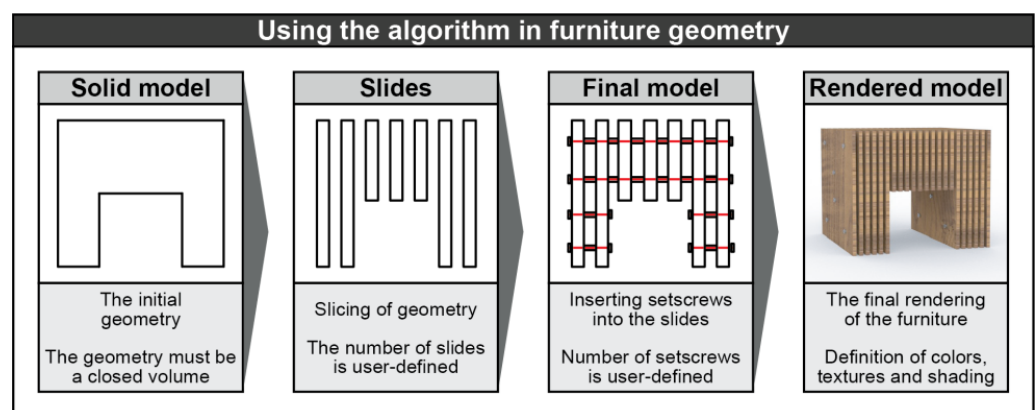


Figure 4. Setscrew design when geometrical gaps appear.

With an aim to support the minimum amount of wooden material used, a nesting feature has been introduced. The present study used a free, open-source Grasshopper 3D™ plugin called OpenNest™. The algorithm automatically derived the nesting position of all parts of the furniture, when the dimension of the plywood sheet was selected. The user could introduce a variety of plywood sheet dimensions that are available for use in industry, while at the same time, the texture properties were applied (Figure 5).

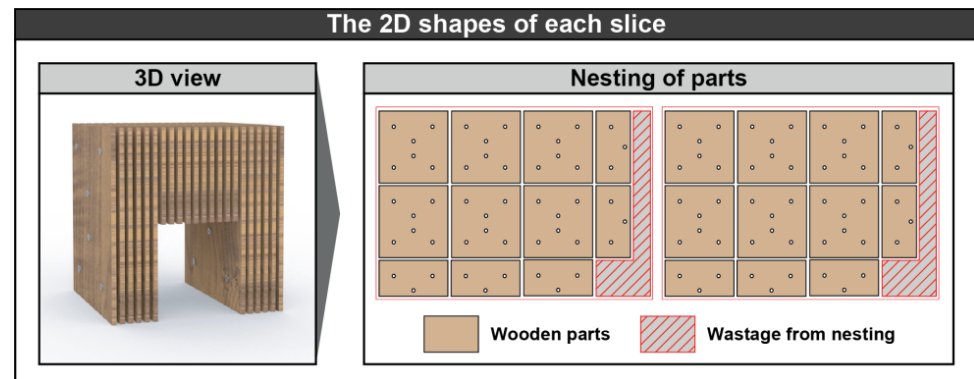


Figure 5. Nesting facility introduced for optimized material usage.

3.4. Output Data

In addition to the nesting feature, the algorithm also supports the calculation of the appropriate coordinates and sizes for the parts, with added functionality to export the required G-code for CNC machining of the wooden components. Additional information is needed on the machinery used in the cutting process, including the equipment manufacturer, digital controller, manufacturing parameters, and tooling. Once this data is inputted, the customized machine program is automatically generated and saved as a text file, ready for use with the available machinery.

A complete Bill of Materials (BOM) is also exported, listing all parts involved in the furniture design, along with their quantities and properties. The BOM includes component volumes and detailed weight calculations. STL and STEP files are exported for prototyping (e.g., 3D printing) and CAD data transfer to other CAD systems. Finally, a set of vector files for all plywood components is exported for various applications, including prototyping and production using laser cutting machinery (see Figure 6).

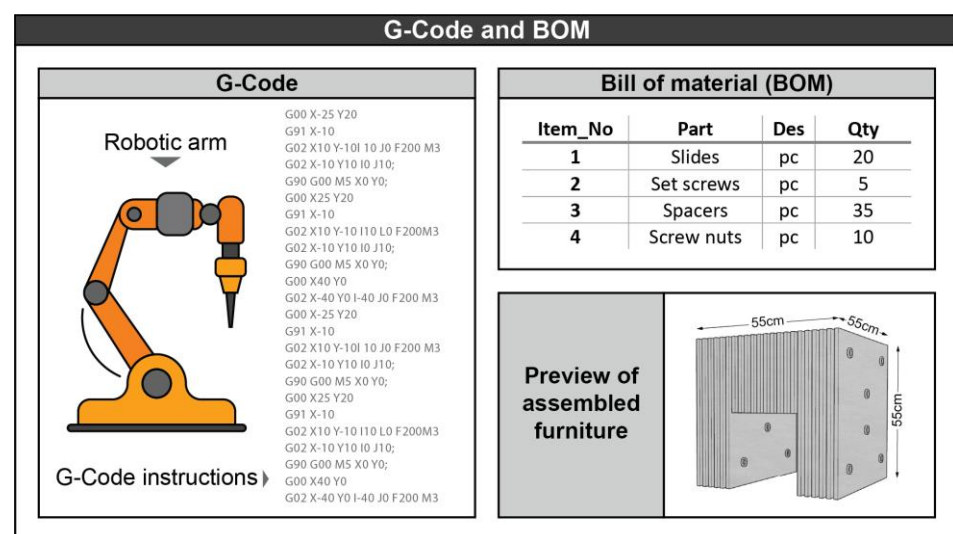


Figure 6. Exported data from the design platform.

4. Case Study

The design platform can accurately handle even unusual geometries and automatically generate all the data described earlier. Figure 6 shows a complex geometry intended to become a bench. A set of input parameters were provided to the platform, and the wooden bench was automatically designed. The furniture dimensions were included in the imported initial file, which contained the geometric details. Other inputs, such as the number of slices, material thickness, setscrew size and path, and material texture, guided the design algorithm. The geometry was sliced, and both spacers and end pieces were calculated and incorporated into the final assembly. Given the furniture's varied shapes, the setscrews were placed with appropriate gaps along their path. The selected materials resulted in a high-quality visual output, appealing to the end user. Choosing alternative textures allowed for several product variations, helping both the designer and the user make informed decisions about the final material selection (see Figure 7).

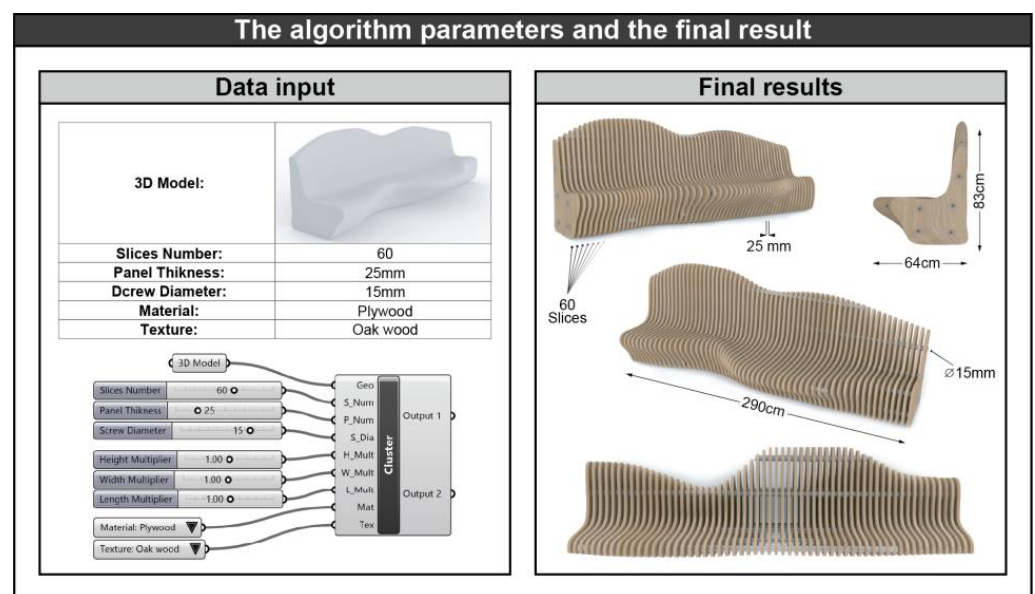
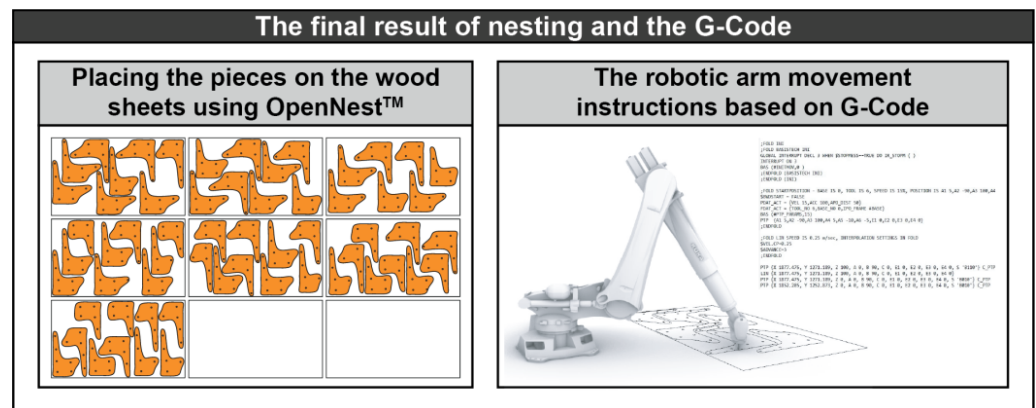


Figure 7. The main input parameters and the resulting furniture design.

All the designed slices were processed to generate the necessary files for CNC machining. First, the slice outlines were used to produce the G-code, which guides a robotic arm (or a Computerized Numerically Controlled CNC machining center) to manufacture the pieces sequentially. Simultaneously, the material usage was minimized by employing an optimized nesting algorithm to reduce plywood demand. Figure 8 shows the nesting result of the slices, achieving 64.4% material efficiency and requiring only 7 wooden sheets instead of the 9 sheets initially calculated manually. Additionally, a fixed orientation constraint was applied, ensuring that all slices had a consistent direction in the wood texture.

The furniture design platform also exports additional information, including technical drawings of all components involved. This supports the designer in organizing the machining process and in managing the purchase of all necessary parts. The data can also be used for costing and warehousing purposes. A 3D printing feature has been integrated to produce physical prototypes for evaluating the final furniture design. The 3D-printed models primarily help with visualizing the shape and proportions of the furniture, as well as assessing the proposed esthetic solution early in the design process. The small size of the prototype (limited by the capabilities of the available 3D printer) allows both the designer and the user to review the design at a low manufacturing cost. In this study, a CREABOT™ D600 3D printer was used to create a relatively large prototype measuring

600 mm in length. In the furniture industry, companies use various CAD software tools, leading to strong demand for neutral file formats that can be universally transferred. For this reason, the platform exports parts and assemblies in multiple CAD formats, including DXF, DWG, STL, OBJ, STEP, and Parasolid. This ensures seamless communication across different companies, especially for manufacturing purposes (see Figure 9). Figure 9 also presents visual assembly instructions for technicians, along with two alternative design solutions beyond the main one presented in the study. In all cases, the platform operated successfully and produced the final furniture without issues.



user can change the initial geometric form and input a very complex geometry, ultimately acquiring all the information for prototyping, manufacturing, and 3D CAD model transfer files demanded by alternative suppliers.

The whole process presented here highlights the new roles that both the furniture designer and the final user must adopt. The designer becomes a skilled CAD programmer in order to automate the design offered. This removes the routine work and encourages design creativity but demands that modern furniture designers have visual and textual programming skills, which is not very common nowadays. From the user's point of view, it is relatively easy to use the designed platforms and see the product's renders and 3D prototypes before deciding on the furniture purchase. This results in increased customer satisfaction and offers improved marketing tools. Finally, the downstream applications lead to improved data transfer among the enterprises involved, i.e., design, tier suppliers, and manufacturing facilities.

The computational time of the design platform refers to the time required by a specific computer to successfully execute the algorithm. This time is heavily influenced by the hardware used and the complexity of the algorithm. In the present study, the platform required approximately one minute to complete the process on a computer with average performance. For comparison, designing the same piece of furniture using a traditional approach would take approximately 8.5 h. This estimate was made as part of an early evaluation by two experienced furniture designers, who manually created the assembled furniture without the use of parametric or automated tools. However, in the case of the computational approach, the time required to initially develop the design platform should also be considered. Table 1 presents the total time required for designing one and ten pieces of furniture using both computational and traditional methods. The results clearly demonstrate the advantage of computational design when dealing with complex geometries and multiple variations.

Table 1. Comparison of required design time between traditional and algorithmic methods.

Design method	When Designing Only 1 Furniture		When 10 Furniture Designs Are Implemented	
	Traditional	Algorithmic	Traditional	Algorithmic
Algorithm design	-	20 h		20 h
Computing time	-	1 m	-	10 m
Traditionally design time	8.5 h	-	85 h	-
Total time	8.5 h	20 h 1 min	85 h	20 h 10 min
	Algorithmic		Traditional	
Material waste	Automatic reduction via nesting		Using external tools for nesting	
Design model accuracy	High accuracy		Unpredictable accuracy	
Total cost	Can be calculated during parameter definition		By experience	
Integration into industry	Yes		No	

By using the design platform's built-in nesting functionality, the placement of slices on wooden sheets was optimized, reducing both the number of sheets needed and the amount of wood waste. As a result, fewer but larger leftover pieces remained, which could be reused in other products. In this case study—featuring an unusual geometry—material waste was reduced by 22.2% compared to a manual, empirical method without nesting. It is important to note that a fixed orientation constraint was applied during nesting to ensure that all slices followed the same direction in the wood grain. While this constraint did increase material waste slightly, the nesting algorithm still reduced the number of sheets

required from 9 to 7. This demonstrates a significant advantage of the proposed platform over traditional methods. Additionally, the total cost of the furniture could be estimated if further fabrication and labor parameters were provided.

5. Conclusions

In the furniture design and manufacturing industry, there is a high demand for digital platforms that drive enterprises towards Industry 4.0. Automating the design and manufacturing process, while involving the end user in this process, offers a great number of advantages:

- It dramatically shortens the time required to design the final product;
- Geometries of high complexity can be used effectively in furniture design;
- A number of automated downstream facilities can become available, i.e., 3D CAD part modeling and 3D CAD assembly generation, detailed 2D/3D technical drawings, robotic arm/CNC machining code generation, BOM, data transfer among several CAD/CAM systems;
- Several wooden materials can be used via an extensive database for rendering, prototyping, and nesting purposes;
- It supports the new role of the furniture design engineer as a CAD programming expert and the end user is encouraged to participate in the design process, increasing customer satisfaction.

This design platform stands out from similar solutions by fully automating the process while still allowing designers to modify the default parameters as needed. Additionally, it can support downstream applications, enabling the development of unique, integrated tools tailored to specific manufacturer requirements. The reduction in material waste and the automation of the entire design-to-production process provide a significant advantage in the context of Industry 4.0 implementation. The platform was tested with various geometries and parameter sets and consistently succeeded in generating the final furniture designs. The integration of 3D printing capabilities proved valuable for the early evaluation of furniture esthetics. The authors believe that incorporating automation into the furniture design phase can drive cultural change toward the adoption of Industry 4.0 principles, ultimately leading to lower production costs and greater end-user involvement.

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